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# COMMERCIAL ITEM DESCRIPTION

SPRAYING UNIT, CLEANING-DEICING- ANTI-ICING FLUID, TRUCK MOUNTED

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-S-26110 for all federal agencies.

## 1. SCOPE.

This commercial item description (CID) describes chassis mounted, diesel powered, spraying units, which are capable of deicing, anti-icing, cleaning, and decontaminating the entire range of airframes in the U. S. Air Force's inventory. Two general classifications are herein defined: first, a basic unit, Class A, that can de/anti-ice and clean all but the upper tail sections of C-5 and C-17 airframes; and, second, a spraying unit, Class B, provided with the extended reach necessary to effectively de/anti-ice C-5 and C-17 horizontal stabilizers. Both types of spraying units include boom-mounted enclosed cabs from which de/anti-icing and decontamination operations are performed. The spraying units are capable of the controlled application of heated deicing fluid, deicing fluid delivered via forced air and/or high efficiency nozzles, and unheated anti-icing fluid. An auxiliary power transmission source such as an engine or hydrostatic drive is provided to hydraulically operate all de-icing components including: boom and enclosed cab actuation, deicing-fluid heater motor and fuel pump, and separate deicing and anti-icing pump motors.

## 2. CLASSIFICATION.

The two types of spraying units covered by this CID are classified as follows:

| Class | Elevation of Boom | Minimum Tank Capacity |
|---|---|---|
| A | 42 feet | 1,600 gal |
| B | 75 feet | 2,600 gal |

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3. SALIENT CHARACTERISTICS.

3.1 Spraying unit description. Both Class A and Class B spraying units shall be constructed in accordance with the requirements of SAE ARP 1247, *General Requirements for Aerospace Ground Support Equipment, Motorized and Non-motorized*, SAE ARP 1971, *Aircraft Deicing Vehicle – Self-Propelled, Large Capacity*, and this document. In the event of a conflict between either ARP 1247 or ARP 1971 and the text of this document, this document shall take precedence. Nothing in this document, however, overrides applicable laws and regulations, unless specific exception has been obtained by the procuring agency. Unless otherwise specified, each of the following salient characteristics shall be applicable to both Class A and Class B spraying units. The spraying unit shall be designed for two-person operation or, when specified, one-person operation. It shall be maintainable with standard tools and built with modular parts supplied from commercial sources.

3.1.1 Modified commercial product. The spraying unit, at a minimum, shall be the manufacturer's standard commercial product modified as necessary to be in accordance with the requirements specified herein. Additional or complimentary features which are not specifically prohibited by these specifications, but which are part of the manufacturer's standard product, shall be included in the unit being furnished.

3.1.2 Components: The spraying unit shall consist of the following major components:

| Item or System Name | See Characteristic |
|---|---|
| Hydraulic system | 3.11 |
| Chassis | 3.12 |
| Truck cab | 3.13 |
| Unit frame | 3.14 |
| Boom assembly | 3.15 |
| Stabilizers | 3.16 |
| Enclosed cab | 3.17 |
| Auxiliary power source | 3.18 |
| Plumbing | 3.19 |
| Pumping systems | 3.20 |
| Fluid heating system | 3.21 |
| Fluid tanks | 3.22 |
| Delivery system | 3.23 |

3.1.3 Service life. The spraying unit shall be designed to have at minimum a service life of twelve years.

3.2 Design and construction. The spraying unit shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the European Conformity, "CE" marking affixed (see 3.2.3.1). It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other



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conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The controls shall be readily operable by personnel wearing heavy, arctic gloves and clothing, or Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Fasteners, handles, and fittings used in the assembly of the item shall also be primed and painted.

3.2.1.1.1 Overseas coating. When specified, (see 7.2) units shall be treated to prevent corrosion from heavy salt water environment when being used aboard ships.

3.2.1.2 Finish. Exterior surfaces shall be prepared, primed, and painted with poly-urethane paint. The standard paint shall be resistant to aircraft hydraulic fluid. Unless otherwise specified (see 7.2), the exterior finish color of the spraying unit shall be Semigloss Green, Color Number 24052 of FED-STD-595.

3.2.1.3 Exclusion of water. The design of the spraying unit shall be such as to prevent water leaking into, or being driven into, any part of the spraying unit interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the spraying unit that could collect and retain water.

3.2.1.3.1 Fluid traps and faying surfaces. There shall be no fluid traps on the spraying unit. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.3.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.3.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the unit is on a one degree incline in any plane. The minimum size of the drain holes shall be 0.25 inch.



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3.2.1.4 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of metals of dissimilar base to provide electromotive compatible abutting surfaces shall be permitted. The use of dissimilar metals separated by suitable insulating material shall be permitted.

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½-inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of FED-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of FED-STD-595. The center of gravity of the spraying unit shall be stenciled on the unit within 1.0 inch of the calculated center of gravity.

3.2.3 Identification and information plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the spraying unit in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The spraying unit and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings. The "CE" marking shall be affixed in accordance with EU requirements on or adjacent to the identification plate.

3.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the spraying unit in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the spraying unit.

b. Horizontal and vertical location of the center of gravity of the spraying unit in air transportable configurations for C-17 and C-5 aircraft, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.



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g. Tie down information.

3.2.4 Safety. In addition to the safety requirements specified in 3.8, 3.9, and 3.10 of SAE ARP 1247 the following safety features shall be included:

a. Emergency shutdown control switches shall be provided in the enclosed cab, truck cab control panel, and by the ground control station to shut down the auxiliary power source fluid and heating system as well as stop all boom movement. The communication system (see 3.17.1) shall not be interrupted by the activation of this switch.

b. Automatic shutdown of pumps in the event of low fluid level.

c. A fire extinguisher system, which includes a mechanism that shuts down the fluid heater upon activation. The system being installed must be certified by the manufacturer for its intended use. The system shall be automatically activated as well as manually operated, as per 3.9.9 of SAE ARP 1971. This system shall have a minimum inspection requirement of "annual" and no special certification should be needed for the inspection of the system.

d. Shutdown devices with manual reset switches in the event of overheating or misfiring of the fluid heater.

e. Automatic 45-second fresh air purge of heater before ignition and manual 150-second fresh air post-purge after heater shut down if the unit contains an air compressor.

f. Interlock system to prevent driver from operating in excess of 5 mph while the boom is in raised, out of boom rest, or cradle position.

g. Means of regaining control of the enclosed cab and boom from ground level in the event of disabling accident or injury to operator in the enclosed cabin by way of a 12-volt (DC) emergency hydraulic pump.

h. Redundant proximity switches shall be provided that identify maximum boom extension and when stabilizers are required in the lower envelope (Class B only).

i. Redundant proximity switches shall be provided that identify minimum boom angle in the upper envelope (Class B only).

j. Audible backup warning alarm.

k. Audible stabilizer deployment alarm (Class B only), which may be the same as the back-up alarm. This alarm should be located at ground level to be heard by personnel in close proximity.



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l. A red or amber flasher, meeting minimum aviation requirements (FAA AC 150/5210-5D), shall be installed on the rotator assembly at the highest and most visible point. A flasher control switch shall be located on the truck cab control panel.

m. Warning decals, as appropriate.

n. Chassis breaks shall lock out in an emergency, with or without power

3.2.4.1 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.2 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the spraying unit with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.2.4.3 Sound levels. The spraying unit sound levels shall conform to the recommended practice indicated in SAE J 1096.

3.2.4.4 Upper deck access. When specified, (see 7.2) access shall be designed in accordance with applicable sections of Operational Safety and Health Administration (OSHA) 29 CFR 1910 in addition to safety requirements listed in 3.2.4 of this document.

a. Upper deck access of the vehicle for maintenance purposes shall be provided.

b. Two Fall Prevention Anchorage points with a minimum capacity rating of 5,000lbs each shall be provided. The inside measurement or diameter of the anchorage point shall be a minimum of 1 inch. Location of anchorage points shall allow for uninhibited safe access to upper deck components and must be approved by AFLCMC/WNZEA. Specifics found in OSHA 29 CFR 1910 .66 App C.

c. A guard for the auxiliary engine harmonic balancer and pulley assembly shall be provided to prevent personnel from coming into contact with rotating and/or reciprocating parts.

3.2.5 Electromagnetic interference (EMI). The spraying unit shall be in accordance with the following conducted and radiated emission and susceptibility requirements of MIL-STD-461: CE102, CE106, CS101, CS103, CS104, CS105, CS114, CS115, CS116, RE102, and RS103.



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3.2.6 Human engineering. The spraying unit shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear.

3.2.7 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

| Material | Minimum Thread Engagement |
|---|---|
| Steel | 1.0 times the nominal fastener diameter |
| Cast iron, brass, or bronze | 1.5 times the nominal fastener diameter |
| Aluminum, zinc, or plastic | 2.0 times the nominal fastener diameter |

3.2.8 Welders and welding. All welders shall be certified to weld in accordance with AWS D1.1 and AWS D1.2, as applicable. The contractor shall make available to the Government certifications for all welders being utilized on the spraying unit. Welding procedures and all welding on the spraying unit shall be in accordance with AWS D1.1 and AWS D1.2, as applicable. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The spraying unit shall be capable of operating in ambient temperatures ranging from -40° F to 130° F.

3.3.2 Storage temperature range. The spraying unit shall be capable of being stored in ambient temperatures ranging from -40° F to 130° F.

3.3.3 Precipitation.

3.3.3.1 Rain. The spraying unit shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the spraying unit at angles from vertical to 45°.

3.3.3.2 Snow. The spraying unit shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours.



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3.3.3.3 Ice. The spraying unit shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

3.3.4 Solar radiation. The spraying unit shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

3.3.5 Fungus. All materials used in the spraying unit shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part.

3.3.6 Salt fog. The spraying unit shall be capable of storage and operation in high temperature, high humidity, salt laden, sea coast environments without damage or deterioration of performance.

3.3.7 Sand and dust. The spraying unit shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

3.4 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 3.5.2) shall not exceed:

| | |
|---|---|
| Weight | 60,000 pounds |
| Length | 496 inches |
| Width | 102 inches (windows retracted) |
| Height | 153 inches |

3.5 Transportability.

3.5.1 Surface transportability. The spraying unit shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

3.5.2 Air transportability. The spraying unit shall be transportable on C-5 and C-17 cargo aircraft in accordance with MIL-STD-1791. The configuration of the spraying unit for air transport may be reconfigured (parts folded in, down, or etc.) but may not be removed. In all air transport configurations, the spraying unit shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The spraying unit shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The spraying unit shall roll on and off the aircraft, negotiating the required maximum ramp angles without shoring

3.5.2.1 Shoring. The spraying unit shall be air transportable without shoring.



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3.5.2.2 Axle weight. Axle weight shall not exceed 20,000 pounds.

3.5.2.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

3.5.2.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 2 man-hours and restoration to operating configuration shall take no more than 2 man-hours for 2 persons using common hand tools (see 7.3.1). All equipment removed shall be stored on the spraying unit; caps and plugs shall permit moving and storage in transport configuration.

3.5.3 Tie downs. The spraying unit shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the spraying unit. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.6 Maintainability. The spraying unit shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females. The spraying unit shall be designed and constructed as follows:

a. There shall be a minimum number of parts consistent with reliability and performance specified herein.

b. All de/anti-icing and decontamination components shall be readily accessible, either by hinged compartment doors or removable panels, which shall not require more than five (5) minutes to open or remove.

c. The spraying unit shall allow adjustments, servicing, maintenance, and replacement of parts and components with minimum disturbance to other equipment, parts, or components.

d. Parts and components shall be positioned for access and ample working space shall be provided, unless performance or reliability is appreciably degraded by the access location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be provided primary access.

e. The number of tools required to accomplish maintenance shall be minimized by applying design practices such as reducing the variety of bolt head sizes.

f. Where possible, all wiring shall be incorporated into harnesses and replaceable within 30 minutes.

g. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.



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3.6.1 Preventive maintenance. The recommended preventive maintenance interval (PMI) shall be at least 200 operating hours, or 8 weeks, whichever occurs first. Daily preventive maintenance tasks shall not require more than 1 man-hour. Other important maintenance intervals include weekly, 250 hour/monthly, and seasonal maintenance checks. The weekly preventative maintenance tasks shall not require more than 2 man-hours and the 250 hour/monthly and seasonal preventative maintenance tasks shall not require more than 8 man-hours.

3.6.2 Inspection and servicing provisions.

a. Routine servicing tasks and pre-use inspections shall require no hand tools.

b. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

c. The spraying unit shall be designed with maximum usage of sealed lifetime lubrication bearings.

d. The spraying unit shall be designed so the correct oil and coolant levels can be visually checked while the unit is running.

3.6.3 Special tools. The design of the item shall minimize the requirement for special tools (see 7.3.2). All special tools shall be provided with, and stored on, the spraying unit.

3.6.4 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the spraying unit shall be provided with each spraying unit on CD-ROM or DVD-ROM.

3.6.5 Special test equipment and software. When specified (see 7.2) the spraying unit shall come shipped with any software or equipment needed to perform any testing not able to be completed with conventional software or common hand tool (see 7.3.1).

3.7 Performance. The spraying unit shall be capable of providing full deicing and anti-icing operations under the following conditions without detrimental effect to subsequent operation:

a. Elevations up to 6,000 feet above sea level.

b. A relative humidity of at least 100 percent at an operating temperature of 130°F.

c. Wind loads of at least seven (7) pounds per square foot.

d. Continuous 25 miles per hour (mph) operating speed on horizontally level surfaces with fluid tanks full in accordance with 3.2.2 of SAE ARP 1971.



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e. Continuous 4 miles per hour operating speed when boom of Class A spraying unit is not cradled or when boom of Class B spraying unit is not cradled and outriggers/ stabilizers are not deployed.

3.8 Mobility. The spraying unit shall be in accordance with SAE ARP 1247, except as otherwise specified herein.

3.8.1 Terrain.

3.8.1.1 Paved surfaces. The spraying unit shall be drivable at speeds up to 20.0 miles per hour (mph) over paved surfaces.

3.8.1.2 Graded gravel. The spraying unit shall be drivable at speeds up to 10.0 mph over graded gravel.

3.8.1.3 Belgian block. The spraying unit shall be drivable at speeds up to 8.0 mph over Belgian block.

3.8.2 Towing force. The towing force shall be in accordance with 3.13.1.3.15of SAE ARP1247.

3.8.3 Clearances. Ground clearance, ramp break-over angle, and angles of approach and departure shall be in accordance with 3.13.1.7.1.8of SAE ARP1247.

3.8.4 Suspension. The spraying unit shall have a suspension system in accordance with 3.13.1.7.1.10 of SAE ARP1247.

3.8.5 Steering. The spraying unit shall have a steering system in accordance with 3.13.1.7.1.15of SAE ARP1247.

3.8.6 Brakes.

3.8.6.1 Parking brakes. The spraying unit shall have parking brakes in accordance with 3.13.1.7.2.3 of SAE ARP1247.

3.8.6.2 Service brakes. The spraying unit shall have service brakes in accordance with 3.13.1.7.2.5 of SAE ARP1247.

3.8.7 Bumper. The spraying unit shall have a rear bumper in accordance with ANSI B56.1

3.8.8 Fenders. The spraying unit shall have fenders in accordance with 3.13.1.7.1.16 of SAE ARP1247.

3.8.9 Mud-flaps. The spraying unit shall have mud-flaps in accordance with 3.13.1.7.1.16 of SAE ARP1247



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3.8.10 Tires and wheels. The spraying unit shall be equipped with single or dual tubeless steel belted radial all-season tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application.

3.9 Engine and related equipment. The spraying unit shall be equipped with a diesel engine. Consistent with the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur. It shall be in accordance with SAE ARP 1052. The Air Force possesses an EPA waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a decal for the engine stating: "This engine has an exemption for national security under 40 CFR 85.1708".

3.9.1 Engine starting system.

3.9.1.1 Starter. The engine shall be equipped with a 12-volt DC electric starter.

3.9.1.2 Engine starting aids. All engines shall start within 15 seconds cranking in any ambient temperature within the required operating range of the spraying unit Internal engine starting aids, fluid starting aids, and heat from the winterization system (see 3.10.3) may be used prior to and during the start period to facilitate engine starting under the following conditions in Table II:

TABLE II. Engine starting aids.

| **Temperature Range** | **Starting Aids Permitted** |
|---|---|
| 40° F through 130° F | None |
| 0° F through 39° F | Internal engine starting aids |
| -40° F through -1° F | Internal engine starting aids and heat from the winterization system |

3.9.2 Engine air intake system. The engine air intake system shall be in accordance with 3.13.1.4.3 of SAE ARP1247. The inlet shall not draw air from directly beneath the spraying unit and shall not be located near the cooling system air outlet nor the engine exhaust outlet. Joints shall be minimized between the air filter outlet and the actual engine air inlet and shall be designed to ensure no leakage of unfiltered air into the engine. A differential pressure air filter service indicator shall be provided.

3.9.3 Engine cooling system. The engine cooling system shall be in accordance with 3.13.1.4.2 of SAE ARP1247. Silicone radiator and heater hoses, constant-torque clamps, a coolant filter, and a coolant recovery system shall be provided. Engine coolant shall be in accordance with A-A-52624, Type I, and of adequate strength to provide protection to -40° F. The engine out (top of radiator) coolant temperature shall not exceed 210° F. The coolant filter shall be rated for one year or 2,000 hours between replacements, with supplementary coolant additives (SCA) either included in the filter or added separately.



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3.9.4 Engine lubrication system. The engine lubrication system shall be designed so that the spraying unit can be operated on a one degree incline in any plane.

3.9.4.1 Engine oil. The engine shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0° F to 250° F. Oil pre-heat for operation below 0° F is allowed. The engine shall be compatible with arctic engine oil in accordance with MIL-PRF-46167 from -40° F to 60° F.

3.9.4.1.1 Engine oil operating temperature. The engine oil sump temperature shall not exceed 250° F.

3.9.4.1.2 Engine oil consumption. The engine oil consumption shall not exceed 0.0035 pounds per brake horsepower-hour (lbs/bhp-hr) under any operating condition.

3.9.4.2 Engine oil filter. The engine oil filter shall be in accordance with 3.13.1.4.4 of SAE ARP1247.

3.9.5 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake, and shall not be directed toward the ground.

3.9.6 Engine fuels and fuel system.

3.9.6.1 Engine operating fuels. The following shall be standard operating fuels:

a. 1-D S15, 1-D S500, 1-D S5000 in accordance with A-A-52557 and ASTM D975, below 60° F ambient air temperature.

b. 2-D S15, 2-D S500, 2-D S5000 in accordance with A-A-52557 and ASTM D975, above 32° F ambient air temperature.

c. JP-5, in accordance with MIL-DTL-5624, -40° F to 130° F ambient air temperature.

d. JP-8, in accordance with MIL-DTL-83133, -40° F to 130° F ambient air temperature.

e. TS-1 in accordance with GOST 10227-86 with JP-8 additives, -40° F to 130° F ambient air temperature.

3.9.6.2 Engine emergency fuels. The following shall be emergency fuels:

Jet A, Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note:

Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.



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3.9.6.3 Engine fuel consumption. When operating on JP-8, the engine at full rated load shall not exceed 0.8 lbs/bhp-hr brake specific fuel consumption.

3.9.6.4 Fuel system. The fuel system shall be in accordance with 3.13.1.5.1 through 3.13.1.5.11 of SAE ARP1247 except as otherwise specified herein. The fuel system shall be constructed of materials which are compatible with the fuels listed in 3.9.6.1 and 3.9.6.2. Copper shall not be used in the fuel system. The fuel system shall be equipped with a fuel shut-off valve(s) to prevent continuous spillage when fuel lines are disconnected for service.

3.9.6.4.1 Fuel priming pump. The spraying unit shall be equipped with an electric fuel pump in addition to the mechanical fuel pump. The electric pump shall be used as a priming pump capable of re-priming the engine fuel system following fuel exhaustion.

3.9.6.4.2 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic using nothing more than common hand tools (see 7.3.1) without loss of engine prime.

3.9.6.4.3 Fuel tank. The fuel tank shall be in accordance with 3.13.1.5.5 through 3.13.1.5.9 of SAE ARP1247. The tank shall be designed so that the spraying unit can be operated on a one degree incline in any plane. The tank shall be provided with corrosion protection and baffles. A 0.25- to 0.375-inch nominal drain valve shall be provided for emptying fuel and sediment into a container underneath the spraying unit without removal of the tank or any other major component. The fuel tank shall have a fuel fill opening of not less than three inch inside diameter and shall be designed to drain fuel spillage overboard for collection outside the spraying unit. The fuel cap shall be equipped with a retention device to prevent loss and FOD. The fuel fill opening, fuel cap, and fuel cap retention device shall be fabricated from non-sparking material.

3.9.7 Engine diagnostic and emergency shutdown systems.

3.9.7.1 Engine diagnostic system. If the engine is equipped with an electronic control module, a diagnostic system shall be provided with a means to indicate engine faults; it shall be equipped with a Controller Area Network (CAN) buss connector. If the spraying unit is equipped with a diagnostic or built-in-test system, the engine diagnostic system shall be integrated with it; if not, it shall be a stand alone system.

3.9.7.2 Engine emergency shutdown system. The engine shall be equipped with an engine emergency shutdown system consisting of a fuel cutoff solenoid activated by the following conditions:

a. Low oil pressure, less than 10 psi or in accordance with the engine manufacturer's recommendations.



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b. Coolant over heat or cylinder head temperature over heat condition in accordance with the engine manufacturer's recommendations.

c. Engine over speed should the engine exceed 110% rated speed. This is required for engines over 100 bhp and optional for engines less than 100 bhp rated horsepower.

3.9.8 Engine operator instruments. The following instruments shall be provided to the engine operator:

a. Tachometer.

b. Coolant temperature gauge.

c. Low coolant level indicator.

d. Oil pressure gauge.

e. Oil temperature gauge.

f. Fuel level gauge.

g. Hour meter.

3.10 Electrical system. The spraying unit shall have a 12-volt, negative ground electrical system in accordance with 3.13.1.2 of SAE ARP1247 except as otherwise specified herein.

3.10.1 Alternator. A single or dual alternator charging system in accordance with 3.13.1.4.9 of SAE ARP1247 shall be provided. The alternator shall be capable of restoring the energy expended during an engine start in less then 15 minutes of engine idle at -40° F.

3.10.2 Batteries and battery compartment.

3.10.2.1 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

3.10.2.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

3.10.2.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings, and shall not be spliced.

3.10.2.4 Booster receptacle. The unit shall come equipped with a booster slaver receptacle that provides the ability to jump using 12 volt battery power from one deicer to another if that other



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deicer also has a similar connector. External lugs shall also be installed in the event another deicer is unavailable.

3.10.3 Winterization system. A winterization system may be provided for starting in temperatures to -40° F. The winterization system may include heaters for engine coolant, engine oil, and the fuel tank, as well as battery warmers. The winterization system shall be designed to operate from an external 220-volt AC power source utilizing the external electrical connections (see 3.10.2.4). A 25-foot cable shall be provided. A visible or audible warning device shall notify the driver if the truck remains connected to the cable when the truck engine is started. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. Any harness will be shipped with the vehicle for electrical connections to operate any winterization system.

3.11 Hydraulic system. The hydraulic system shall be in accordance with 3.13.1.3 of SAE ARP1247 except as otherwise specified herein. O-ring face seal hydraulic fittings may be used in lieu of flared fittings (see 3.13.1.3.12 of SAE ARP1247). Hydraulic fluid shall be in accordance with MIL-PRF-83282. All hydraulic system components, including the hydraulic tank, shall comply with all corrosion resistance requirements specified herein. The system shall be equipped with a hydraulic oil cooler and thermostatically controlled fan to maintain the hydraulic oil at the appropriate temperature to ensure proper operation of the unit within the operation envelope specified within this document. A low-level shutdown system shall be provided to protect the hydraulic system. The hydraulic oil reservoir shall be equipped with an externally mounted level and temperature gauge. External manual controls shall be provided in the event of electrical solenoid failure.

3.11.1 Hydraulic pressure test system. Test ports with quick disconnects shall be provided, along with a hydraulic test gauge, to allow a mechanic to quickly test for correct pressure in the various hydraulic circuits.

3.11.2 Electric-driven emergency hydraulic system. An emergency hydraulic system, driven from the 12-Volt DC battery, shall be provided. This system shall enable the aerial device to be maneuvered and lowered in the event of auxiliary power supply failure.

3.12 Chassis. A commercial or custom truck chassis in accordance with 3.3 of SAE ARP 1971, which has been modified to satisfy the requirements of either a Class A or Class B spraying unit configuration, shall be provided. The chassis shall interface with the unit frame, boom assembly, and all other components in such a manner that the entire assembly exists as an integrated whole.

3.12.1 Transmission. The driveline shall include an automatic transmission. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

3.12.2 Tow hooks. Tow hooks shall be installed on the chassis frame, or on suitable members attached directly to the chassis. There shall be two located in both, the front and rear, one on each side of the centerline. They shall be in compliance with the requirements in 3.8.2.



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3.13 Truck cab. The chassis shall include a standard truck cab with seating for vehicle driver and one passenger. The driver's position shall be on the left side of the chassis to provide the maximum possible view of the enclosed cab operator and enclosed cab under all operating conditions. Power and heated mirrors shall be included.

3.13.1 Truck cab control panel. In addition to the standard instrument panel provided by the chassis manufacturer, a separate control panel for deicing operations shall be provided that includes the following:

a. Auxiliary power supply controls (see 3.18.2).

b. Fluid pump on/off switch(es).

c. Heater on/off switch and activation indicator light.

d. Flashing (red or amber) beacon on/off switch.

e. Emergency stop switch (see 3.2.4-a.) and activation indicator light.

f. Overhead window washer/wiper activation switch.

g. Gauge lights for nighttime operation.

h. Deicing & anti-icing fluid level gauges.

i. Deicing fluid pressure gauge.

j. Deicing fluid temperature gauge.

3.13.2 Truck cab observation window. The cab roof shall be fitted with a large, sloped observation window made out of safety glass or similar material. An electric wiper for overhead viewing from either the driver or passenger's seated position shall be included.

3.13.3 Truck cab boom controls. In addition to the enclosed cab boom controls (see 3.17.2), identical boom actuation and cab rotation fully proportional joystick controls shall be installed between the driver and passenger, so that either may conveniently operate them. In accordance with 3.4.8 of SAE ARP 1971, these controls shall override the enclosed cab controls. These controls shall not operate when the boom is in the upper operational envelope, between 42 feet and 75 feet of elevation, of the Class B spraying unit.

3.14 Unit frame. All components of the deicing unit, including fluid tanks, auxiliary engine, heater, fluid pumps, and boom pedestal shall be mounted on a frame that fastens to the truck chassis. This framework shall be designed to allow for mounting on any chassis that conforms to the manufacturer's requirements and containing all the relevant modifications for either a Class A or Class B configuration.



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3.15 Boom assembly. The boom assembly or aerial device shall be in accordance with ANSI A92.2. It shall be specifically designed to achieve the maximum in maneuverability, stability, and safety for de/anti-icing operations within the specified range of either the Class A or Class B spraying unit. Under all operating conditions, the boom shall operate smoothly. The boom shall rotate 360 degrees, non-continuously in accordance with 3.4.3 of SAE ARP 1971.

3.15.1 Boom stability – Class A. The boom assembly shall be designed to support a working load of at least 350 pounds in the enclosed cab during mobile operations or while in any extended position. With a 700-pound load placed in the enclosed cab with the boom assembly in any extended position, with no fluid in the de/anti-icing fluid tanks, the deicing unit on level ground shall be capable of remaining stable and not overturn.

3.15.2 Boom stability – Class B. The boom assembly shall be designed to support a working load of at least 250 pounds in the enclosed cab during mobile operations or while in any extended position. With a 500-pound load placed in the enclosed cab with the boom assembly in any extended position, with no fluid in the de/anti-icing fluid tanks, the deicing unit on level ground shall be capable of remaining stable and not overturn.

3.15.3 Range of platform movement – Class A. The boom assembly of the Class A spraying unit shall raise the spray nozzle to a minimum vertical height of 42 feet as measured from level ground to the floor of the enclosed cab. The boom assembly shall also have a horizontal, side reach of not less than 40 feet as measured from the fluid spray nozzle to the centerline of the turntable.

3.15.4 Range of platform movement – Class B. In addition to the range of platform movement for a Class A spraying unit, as specified in 3.15.3, the boom assembly of the Class B spraying unit shall raise the spray nozzle to a minimum vertical height of 75 feet as measured from the level ground to the floor of the enclosed cab. The boom assembly shall also have a horizontal, side reach of not less than 40 feet as measured from the fluid spray nozzle to the centerline of the turntable.

3.15.5 Operational envelope/limit indicators – Class B. Situated near or within view of any set of boom controls, the Class B spraying unit shall be equipped with a light panel that indicates via proximity sensors whether the enclosed cab is in the lower (below 42 feet) operational envelope or the upper (42 feet or above) operational envelope. Each light panel, along with audible warning, shall also provide indication when the boom has reached or is in proximity to any of its operational limits, such as maximum/minimum angle, maximum extension, and the full extent of the lower envelope, signifying that stabilizer deployment is required for further extension. This system shall only allow the boom to operate in stable configurations. All audible warnings shall be clearly heard in both the enclosed and truck cabs under all operating conditions.

3.15.6 Boom controls. In addition to the enclosed cab (see 3.17.2) and truck cab (see 3.13.3) controls, hydraulic boom controls shall be provided at a ground level control station. The ground level boom controls shall be fully proportional and shall extend/retract and rotate



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horizontally/vertically the boom. These ground level controls shall override either the enclosed cab or truck cab boom controls.

3.16 Stabilizers. If the Class B deicing unit is equipped with outriggers or stabilizers, the unit shall be in accordance with 3.10.9 of SAE ARP 1247, which shall be engaged whenever the enclosed cab is in the upper operational envelope, from 42 feet to 75 feet in elevation. Electric over hydraulic controls for the outriggers or stabilizers shall include the following interlocked features to assure safe operation:

a. Before the stabilizers can be deployed the truck transmission shall be in neutral and the parking brake set.

b. Before the boom can be extended into the upper operational envelope (above 42 feet), the stabilizers shall be fully deployed.

c. Before the stabilizers can be retracted the boom shall be in the lower envelope.

d. Before the truck transmission can be placed in gear the stabilizers shall be completely stowed. Means to override this interlock shall be provided, for emergency situations when the truck needs to be driven and completely stowing the stabilizers is not possible.

e. Cross-port check valves shall be installed to prevent hydraulic drift of each stabilizer.

3.17 Enclosed cab. An enclosed cabin along the middle to upper portion of the boom assembly shall be provided, in full accordance with SAE ARP 5058, "Enclosed Operator's Cabin for Aircraft Ground Deicing Equipment." The enclosed cab shall have sufficient room for at minimum, one operator. The enclosed cab shall come equipped with externally mounted side mirrors to increase operator side and rear visibility during cab maneuvering. Means to automatically level the enclosed cab, while the boom is anywhere within its operational range, shall be included. The enclosed cab shall be capable of horizontal rotation of a minimum of 180 degrees in either direction from straightforward in the plane of the boom. All parts of the enclosed cab that may come in contact with the aircraft under normal operation shall be padded with a resilient, replaceable bumper.

3.17.1 Communication system. A means for a truck cab passenger, the truck driver, and enclosed cab operator to communicate simultaneously shall be provided. The system shall be powered with the on/off/start switch of the truck chassis. If a two-way headset communication system is utilized it shall be in accordance with 3.3.9 of SAE ARP 5058 and 3.4.17 of SAE ARP 1971.

3.17.2 Enclosed cab controls. Easily operated, readily accessible, and consolidated controls for boom actuation, cab rotation, de/anti-icing fluid operations, and forced air operations shall be provided inside the enclosed cab. Controls shall be fully proportional, electric over hydraulic, and in accordance with 3.3.5 of SAE ARP 5058 and 3.10.6 of SAE ARP 1247D, of the "dead-man" type. Variable deicing fluid flow shall be controllable, while non-variable anti-icing flow,



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to protect the fluid from degradation, is recommended. Controls for wipers, washers, emergency stop, and emergency hydraulics shall be conveniently located.

3.17.3 Anti-contact. When specified, (see 7.2) an operator aid system consisting of two (2) side mounted physical antenna sensors that deploy when the boom is raised from the boom rest and a nozzle mounted antenna sensor. If any one of the three (3) antenna sensors contact an object causing the antenna's to deflect then all boom and cab movement functions are immediately stopped. The operator must use the manual override system to move away from the object that is contacting the antenna system.

3.18 Auxiliary power supply. There shall be a means to efficiently drive all de/anti-icing, cleaning, and decontamination components of the spraying unit separate from the main engine.

3.18.1 Auxiliary engine. If a diesel auxiliary engine is utilized, it shall be in accordance with SAE ARP 1052, "Selection Criteria for Internal Combustion Engines Used in Ground Support Equipment," The engine capacity shall be such that the continuous duty horsepower input required by all engine-driven items shall not exceed 75 percent of the full throttle rating of the engine, at sea level and 59° F. The engine shall have a minimum of four cylinders.

3.18.1.1 Auxiliary engine exhaust. The auxiliary engine exhaust shall be routed and properly guarded so as to eliminate any possibility of danger to personnel, components, and wiring. The exhaust shall terminate at the rear of the spraying unit no less than 95 inches above the ground, and shall be equipped with a spark-arresting muffler.

3.18.2 Auxiliary power supply controls. All auxiliary power supply controls shall be mounted on the truck cab control panel. Auxiliary power supply controls, indicators, and gauges shall include: off/on/start switch, fuel level gauge, tachometer, oil pressure gauge, water temperature gauge, battery charge indicator gauge, and hour meter.

3.18.3 Auxiliary power supply fuel tank. In addition to the standard truck chassis fuel tank a separate fuel tank with a minimum capacity of 150 gallons shall be provided. The fuel system shall be capable of operating the auxiliary power supply and heater at rated output for a minimum of four hours without refueling. The fuel system shall be designed to provide for shutdown of the heater first, auxiliary power supply second and chassis engine last. The chassis fuel tank and the 150 gallon fuel tank will be connected together to provide additional capacity.

3.19 Plumbing. The de/anti-icing system shall be designed to be compatible with all types of deicing and anti-icing fluids, as well as, cleaning compound solutions conforming to MILPRF-87937. Plumbing hard lines and fittings shall be non-corrosive and wherever possible stainless steel.

3.20 Pumping systems. Two independently plumbed pumping systems shall be provided to drive heated deicing fluid and unheated anti-icing fluid to the delivery system.

3.20.1 Deicing fluid pump. A pump shall be provided for the delivery of deicing fluid from the deicing fluid tank, through the heater, and to either the spray nozzle of the enclosed cab or the



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ground hand nozzle on the front bumper of the chassis. The pump shall be capable of delivering heated deicing fluid from the enclosed-cab nozzle at a flow rate and pressure sufficient enough to deice the aircraft but not damage the surface.

3.20.2 Anti-icing fluid pump. A pump which will minimize degradation to the anti-icing fluid shall be provided. The pump shall be capable of delivering unheated anti-icing fluid to the spray nozzle of the enclosed cab at a flow rate and pressure sufficient enough to apply fluid but not high enough to degrade (being broken down or diffused enough to be rendered ineffective) the anti-icing fluid. Pump shall shut off when spray nozzle is closed to further ensure limited degradation of the anti-icing fluid.

3.20.3 Fluid refill system. The anti-icing fluid pump system shall include the means to reverse flow, so that the anti-icing fluid tank may be filled from 55-gallon drums with the anti-icing fluid pump providing the necessary suction. The same such system shall be made available when specified for refilling deicing fluid tanks. The systems shall refill the fluid tanks at a minimum rate of 18 gallons per minute and without degradation to the fluids being refilled.

3.21 Fluid heating system. A heater that operates on either diesel fuel, Jet A or JP-8, or other power source shall be provided to heat deicing fluid. It shall have fully automatic ignition and controls for smokeless operation under all conditions, including starting and shut down. The fluid heater system shall have a minimum output heat capacity of 2,000,000 Btu per hour and shall provide maximum nozzle temperature in less than five (5) minutes when the fluid temperature starts at 50° F. It shall be equipped with a thermostat that maintains a fluid temperature of 185° F, plus or minus 10° F, and an automatic shutdown that shall activate when the fluid reaches an excessive temperature.

3.21.1 Heater ignition. The heater ignition shall operate at temperatures as low as -40° F. It shall be in accordance with NFPA 70 and ignition cables within the system shall be in accordance with UL 558.

3.21.2 Combustion air blower. If applicable, a combustion air blower, powered by the auxiliary power supply pump, shall be provided as an integral part of the heater system. This blower, which shall automatically adjust in speed in response to the heater output requirements, shall provide the proper amount of air for fuel/air mixture.

3.21.3 Heater exhaust. The heater exhaust shall be directed up and to the rear of the unit. The exhaust outlet shall be protected from precipitation. It shall terminate no less than 95 inches vertically from the ground. The gas duct shall not be constructed of flexible metal tubing.

3.21.4 Heater control. Electrical controls shall be centrally located at a heater control panel, with the exception of the thermal switches that may be located on the heating unit and the controls mentioned in 3.13.1 of this document. The heater control panel shall include a method of displaying fluid inlet and outlet temperature, fuel pressure, combustion air pressure, and fluid flow. These parameters shall be capable of being monitored when the heater is idling or in operation.



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3.21.5 Heater safety. The following safety features shall be incorporated into the heating system:

a. The ignition system shall prevent discharge of fuel until proper ignition is assured.

b. The heater during operation shall not be adversely affected by vibration, sudden shock, or tilted operation.

c. The heater shall include a flow control switch to shut off fuel flow to the burner when there is no fluid flow through the heater.

d. A safety valve shall automatically relieve pressure in the heater coil when the normal operating pressure is precariously exceeded.

e. The heater shall shut down when exhaust temperatures are excessive.

f. The Heater shall meet PONYA and MASSPORT requirements.

3.22 Fluid tanks. De/anti-icing fluid tanks shall be durably constructed of stainless steel. While the tanks may be of integrated construction, the deicing tank and anti-icing tank shall be physically and thermally isolated. The tanks shall be vented to allow for expansion and shall have internal perforated baffles to prevent fluid impact or sloshing during movement or rapid stopping. They shall also have sufficient-size openings to allow access to a mechanic performing repairs or maintenance as well as an easily accessible fluid sampling port to collect samples of the deice/anti-ice fluids in the tank. The tanks shall be equipped with a sump and holes for intake and return lines. The deicing fluid tank shall also be equipped with a strainer.

3.22.1 Fluid tank capacity – Class A. For Class A spraying units, the total fluid capacity shall be a minimum of 1,600 gallons, in accordance with 3.2.5 of SAE ARP 1971. The deicing fluid tank of the Class A unit shall have a minimum fluid capacity of 1,450 gallons and the anti-icing fluid tank shall have a minimum fluid capacity of 150 gallons.

3.22.2 Fluid tank capacity – Class B. For Class B spraying units, the total fluid capacity shall be a minimum of 2,600 gallons. The deicing fluid tank shall have a minimum capacity of 2,250 gallons, while the anti-icing fluid tank shall have a minimum capacity of 350 gallons.

3.22.3 Tank-level indicators. Level gauges in the truck cab and level sight gauges on the outside of the spraying unit shall be provided. The truck cab also shall be provided with an indicator that registers when the deicing tank is at such a low level that the deicing system has shut down to prevent damage to the deicing pump. Indicator lights activated by an in-tank sensor shall provide the operator with a clear visual signal indicating the deice/anti-ice, and/or water tank is full.

3.22.4 Tank drain handles. Drain handles shall be located to the side of the spraying unit to allow for easy opening and closing of tank drain valves.



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3.23 Delivery system. Operation of the delivery system, consisting of the de/anti-icing fluid spray nozzle, forced air system, and ground handline system shall be independent and not be contingent upon operation of the heater or the boom. All components of the delivery system shall withstand the working pressures specified and a hydrostatic pressure of at least twice the specified working pressure without leakage or failure.

3.23.1 De/anti-icing fluid spray nozzle. A dual fluid spray nozzle shall be capable of variably delivering deicing fluid up to a minimum of 40 gallons per minute with the boom fully elevated. The nozzle shall also be capable of delivering anti-icing fluid up to a minimum of 20 gallons per minute without fluid degradation with the boom fully elevated. Various spray patterns shall be able to be selected from the controls. The spray patterns shall include a wide spread fine mist, a heavy conical spray, and a solid straight stream.

3.23.2 Forced air system. If the manufacturer utilizes a compressor and/or forced air system it shall be in accordance with 3.23.2.1 through 3.23.2.2.1.

3.23.2.1 Forced air nozzle. A separate nozzle to provide forced air, piped from the air compressor shall be provided. The air shall be delivered from the nozzle at a volumetric flow rate of no less than 1,340 cubic feet per minute and at a velocity of 700 miles per hour. The pressure of the air emitted at the nozzle shall not be in excess of 13 pounds per square inch.

3.23.2.2 Forced air system. System shall be manufactured such that it is capable of continuously supplying air to the forced air nozzle. The air outlet shall be such that it will prevent the ingestion or discharge of internal components or damaging fragments in the event of a catastrophic failure of the system.

3.23.2.2.1 Air system meter. An hour-meter shall be provided to allow for monitoring the hours the blower has been in operation.

3.23.2.3 Reduced flow. The system shall include a means to apply deicing fluid in a reduced flow setting for lighter ice loads such as frost at a flow rate of no more than 10 gallons per minute.

3.23.3 Control and range of nozzle motion. A control system within the enclosed cab shall remotely control both the de/anti-icing fluid and forced air nozzles, which may be joined together and move in unison. Both nozzles shall have a minimum horizontal range of movement of 180 degrees to the right and left of the straight-ahead position. The vertical range of movement shall be 90 degrees up and a minimum of 45 degrees down from the horizontal position. The nozzles shall operate at a minimum speed of 5 degrees of arc per second.

3.23.4 Ground handline system. A hose reel shall be mounted to the unit for storing a 50-foot ground delivery hose with a combo-nozzle for quick disconnects on the class A unit and 100-foot on the class B unit. A shut off and drain valve shall be provided to permit draining and removal of the hose reel without interfering with operation of the rest of the system. It shall be capable of retracting the hose unassisted by operators such as through mechanical or electrical force. If



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implemented, the rewind switch shall be adjacent to the hose reel and protected from interference with the hose. The reel should also be equipped with a manual rewind hand crank and brake.

3.23.5 Fluid usage meter. The spraying unit shall be equipped with a fluid usage meter, which can be used to store volumetric history. A display shall be installed in full view of the enclosed cab operator. This display shall be clearly visible under all operating conditions.

3.23.6 Variable fluid blending system. When specified, (see 7.2) units shall have a variable fluid blending system so that glycol percentage in fluid mixture may be adjusted to conditions for optimal application.

3.23.7 Suction fill. When specified, (see 7.2) units shall be capable of filling deicing and anti-icing fluid tanks from the bottom from an external source using suction fill.

3.24 Rear camera. Units shall be equipped with a dual camera video system with a rear-facing camera mounted on the rear of the deicer and the second camera rear of the enclosed cab. The monitor is in the enclosed cab and will automatically switch to the rear camera when the operator moves the joystick to the rear.

3.25 Single operator drive system. When specified, (see 7.2) unit shall be able to drive as well as control all critical deicer functions from the enclosed cab with only a single operator. The deicer can be operated in either single or dual operator modes in this configuration.

3.26 Lighting. All lighting shall have easily procurable replacement bulbs and fixtures.

3.26.1 Floodlights. Exterior non-glare white lighting shall be provided for night operation. Lights shall be positioned to provide adequate working light at ladder(s), access step(s), and work area(s). Lights can be manually activated by operator

3.27 Workmanship. The spraying unit including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.27.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.27.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.



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3.27.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.27.4 Cleaning. The spraying unit shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.28 Product familiarization video. When specified (see 7.2), a product familiarization video shall be provided to aid visually the acclimation of a user to the features present on the spraying unit.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production spraying unit shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5.1 through 5.5.12. The



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contractor shall provide or arrange for all test equipment and facilities. Except as otherwise specified, all testing in which the engine is operated shall be performed using JP-8 turbine fuel.

5.3 Conformance inspection. Each production spraying unit shall be subjected to the examination described in 5.5.1.

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the spraying unit shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Degradation of anti-icing fluid

d. Structural failure of any component, including permanent deformation, or evidence of impending failure.

e. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer’s (OEM’s) specifications or tolerances shall be utilized for making a determination.

f. Evidence of corrosion or deterioration.



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g. Misalignment of components.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Interference between the spraying unit components or between the spraying unit, the ground, and all required obstacles, with the exception of normal contact by the tires.

j. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

k. Overheating of the engine, transmission, or any other vehicle component.

l. Shutdown faults from:

i. Engine cooling system.
ii. Engine lubrication system.
iii. Engine protective circuits.

5.5 Test methods.

5.5.1 Examination of product. Each spraying unit shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each spraying unit function shall be verified. Certifications and analyses shall be provided in accordance with Table III. Each production spraying unit shall be inspected to a Government-approved reduced version of the checklist.



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TABLE III. Certifications and analyses.

| Paragraph | Required Certifications and Analyses |
|---|---|
| 3.2 Design and construction. | Contractor documentation that the spraying unit is in accordance with all applicable EU requirements in order to have the "CE" marking affixed. |
| 3.3.3.2 Snow. | Contractor analysis of the snow load requirement (see 5.5.4.3.2). |
| 3.3.4 Solar radiation. | Contractor certification that the spraying unit performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments. |
| 3.3.5 Fungus. | Contractor certification that the materials used in construction of the spraying unit are fungus resistant or suitably treated to resist fungus. |
| 3.5.1 Surface transportability. | Contractor surface transportability analysis (see 5.5.6.1.1) and certification that the spraying unit is transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, and water transport. |
| 3.5.2 Air transportability., 3.5.2.1 Shoring. | Contractor air transportability analysis (see 5.5.6.2). |
| 3.5.3 Tie downs. | Contractor tie down provision analysis (see 5.5.6.4.1). |
| 3.5.4 Lifting provisions. | Contractor lifting provision analysis (see 5.5.6.5.1). |
| 3.9 Engine and related equipment., 3.9.3 engine cooling system., 3.9.4 Engine lubrication system., 3.9.6.1 Engine operating fuels., and 3.9.6.2 Engine emergency fuels. | Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including cooling system, lubrication system, and mounting system. |

5.5.2 Sound level test. A first production spraying unit shall be tested for compliance to the recommended practice in SAE J 1096 indicated in 3.2.4.3.

5.5.3 Electromagnetic interference test. A first production spraying unit shall be tested in accordance with MIL-STD-461: CE102, CE106, CS101, CS103, CS104, CS105, CS114,



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CS115, CS116, RE102, and RS103 to demonstrate compliance with 3.2.5. Any engine fuel in accordance with 3.9.6.1 may be utilized.

5.5.4 Environmental testing.

5.5.4.1 High temperature storage and operation test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 501.5, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.3.1 and 3.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.4.2 Low temperature storage and operation test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 502.5, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.3.1 and 3.3.2, as well as the engine starting requirement of 3.9.1 and the alternator charging requirement of 3.10.1. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.4.3 Precipitation.

5.5.4.3.1 Rain test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 506.5, Procedure I, to demonstrate compliance with 3.3.3.1.

5.5.4.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 3.3.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810).

5.5.4.3.3 Ice accretion test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 521.3 with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 3.3.3.3. The contractor shall identify those areas of the spraying unit where ice removal is required prior to operation.

5.5.4.4 Salt fog test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 509.5, to demonstrate compliance with 3.3.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

5.5.4.5 Sand and dust test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 510.5, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 3.3.7.

5.5.5 Weight and dimension tests.

5.5.5.1 Weight and center of gravity test. The weight, center of gravity, and axle weights of a first production spraying unit shall be measured to demonstrate compliance with the weight requirement of 3.4 and the axle weight requirement of 3.5.2.2.



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5.5.5.2 Dimension measurement. A first production spraying unit shall be measured to demonstrate compliance with the dimensional requirements of 3.4.

5.5.6 Transportability verification.

5.5.6.1 Surface transportability verification.

5.5.6.1.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 3.5.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, Method 514.6, Table 514.6C-II.

5.5.6.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 3.5.2, 3.5.2.1, and 3.5.2.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.5.2. The evaluation shall also include a dimensional analysis for the spraying unit while traversing the ramp and while loaded aboard C-5 and C-17 aircraft.

5.5.6.3 Equipment removal and reconfiguration demonstration. A first production spraying unit shall be configured for transport on C-5 and C-17 aircraft and then reconfigured for operation to demonstrate compliance with 3.5.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

5.5.6.4 Tie down provision verification.

5.5.6.4.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 3.5.3.

5.5.6.4.2 Tie down provision test. A first production spraying unit shall be tested to demonstrate compliance with the tie down provision requirements of 3.5.3.

5.5.7 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

5.5.8 Mobility tests. A first production spraying unit shall be tested in accordance 4.6.9 of SAE ARP1247.

5.5.9 Brake tests.

5.5.9.1 Parking brake test. The spraying unit shall be tested to demonstrate compliance with 3.13.1.7.2.3 SAE ARP1247.



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5.5.9.2 Service brake test. The spraying unit shall be tested to demonstrate compliance with 3.13.1.7.2 SAE ARP1247

5.5.10 Boom stability test. The boom of the product sample(s) shall be operated throughout its operating range to demonstrate (see 3.15) the stability and smooth operation of the product sample. With 700 pounds placed in the enclosed cab of a class A product sample (if applicable) and 500 pounds placed in the enclosed cab of a class B product sample (if applicable) when on level ground, the boom shall be extended to the maximum position with no fluid in the de/anti-icing fluid tanks; the unit shall remain stable and shall not overturn. Boom certifications shall be included and paper copies should accompany the vehicle prior to shipment. A certification placard shall be affixed to the vehicle.

5.5.11 Pump and fluid delivery test. Manufacturer shall provide test results showing conformance of unit characteristics with all fluid flow values within 3.20.3, 3.23.1, and 3.23.2.3.

5.5.12 Welds and welding. Welds shall be inspected to ensure proper penetration and verified to be in strict accordance with the Visual Inspection criteria set forth in the AWS Codes. The inspection shall be performed by an AWS Certified Weld Inspector (CWI) during and after the construction of all Aircraft Staircases. Documentation of all inspections for the first production Aircraft Staircase shall be provided to the Government. The Government reserves the right to require verification inspection at any time during construction in accordance with 6.1.2.2 of AWS D1.1, 5.1.1 of AWS D1.2, and Non-specified Non-Destructive Test (NDT) other than Visual in accordance with 6.6.5 of AWS D1.1. All welding documentation for subsequent production units shall be made available to the Government at any time during the life of the contract pursuant to AWS D1.1 and D1.2.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents, except for GOST 10227-86, are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094. A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).

7.1.2 FAR. FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/.

7.1.3 ASTM documents. Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959.



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7.1.4 AWS documents. AWS Standards may be obtained at http://www.anweld.org/ or from the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.

7.1.5 SAE documents. SAE Standards may be obtained at www.sae.org or from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.1.6 ANSI documents. ANSI Standards may be obtained at www.ansi.org or from American National Standards Institute, 25 West 43rd Street, New York, NY 10036.

7.1.7 UL documents. UL Standards may be obtained at http://www.ul.com/ or from the Underwriters Laboratories, 333Pfingsten Road, Northbrook, IL 60062-2096.

7.1.8 NFPA documents. NFPA Standards may be obtained at http://www.nfpa.org/ or from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169-7471.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. If overseas transports provisions are required (see 3.2.1.1).

c. Finish color required if not Semigloss Green (see 3.2.1.2).

d. If any special test equipment or software shall accompany the vehicle (see 3.6.5).

e. If upper deck access provisions are required (see 3.2.4.4).

f. If an anti-contact system is required (see 3.17.3)

g. If variable fluid blending system provisions are required (see 3.23.6).

h. If the deicing suction fill is required by the user (see 3.23.7).

i. If a single operator drive system is required by the user (see 3.25).

j. Whether a product familiarization video is required (see 3.28).

7.3 Definitions.

7.3.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.



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7.3.2 Special tool. A tool that is not commercially and readily available from a source other than the spraying unit contractor.

7.4 Key words.

Boom and enclosed cabs
Decontaminating
Extended reach
Heated and unheated

Custodians:
Air Force – 84

Preparing Activity:
Air Force – 84

Reviewer:
Air Force – 99

Agent:
Air Force – 99

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